Determinants of Blood Pressure Variability: A Multilinear Regression Approach

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Introduction

High blood pressure (BP) is a global health crisis associated with life threatening conditions such as heart disease and stroke (Martinez-Ríos et al., 2021). BP is influenced by diverse interconnected factors, understanding them is essential for developing effective prevention strategies. This leads to our research question: **How do physiological and lifestyle factors influence systolic blood pressure variability (BPV)?** This study aims to build upon existing research by examining the combined effects of multiple physiological and lifestyle factors on BPV, extending our understanding on their collective influence.

Past research offers informing insights for our direction of study on blood pressure. Martinez-Ríos et al. (2021) identified a strong positive correlation between systolic BP (SysBP), age and BMI, as well as a higher prevalence of hypertension in men compared to women. Jensen et al. (2024) observed a dose-response relationship between alcohol consumption and BP, with each additional drink per week increasing SysBP by 0.15-0.17 mmHg. Fagard and Cornelissen (2007) demonstrated that endurance training over 4-52 weeks significantly reduced SysBP by 3.3 mmHg. These knowledge directly suggested the predictors of this study: **gender, age, BMI, alcohol drinking frequency, and exercise frequency.**

Linear regression is suitable for our research, as it models BPV as a linear combination of the selected predictors. This allows us to quantify the contribution of each variable while simultaneously assessing their combined impact on the average outcome. By emphasizing interpretability, the model's coefficients reveal both the strength and direction of these associations. This modeling approach will provide meaningful insights into how physiological and lifestyle factors individually and collectively influence BPV.

Methods

BPSysAve was selected as the response variable because it directly measures blood pressure. The preliminary model included BPSysAve as the response and Gender, Age, BMI, AlcoholYear, and PhysActiveDays as predictors, where AlcoholYear represents annual drinking frequency and PhysActiveDays denotes weekly active days.

We began with checking conditions and assumptions on the preliminary model.

1. Condition 1 & 2: We examined Condition 1 (condition mean response) for random diagonal scatter or easily identifiable non-linear trends in the Response vs. Fitted plot, and Condition 2 (conditional mean predictors) for lack of non-linear patterns in pairwise predictor scatterplots. If Condition 1 is violated, then linear regression is not suitable for this research. If Condition 2 is violated, in the subsequent diagnosis on model assumption, we should decide by cases if an assumption is indeed violated.
2. Linearity: Residual plots were checked for systematic patterns, such as curves or functions of predictors, suggesting non-linearity. Box-Cox transformations were applied to address violations.
3. Constant Variance: Fanning patterns in residual plots indicated heteroskedasticity, mitigated using variance-stabilizing transformations.
4. Uncorrelated Errors: Residual plots showing large clusters of many points, a pattern across time, or some sequencing information suggested autocorrelation, though no corrective measures could be pursued in this analysis.
5. Normality: Deviations from the diagonal in Q-Q plots indicated non-normality, corrected using Box-Cox transformations as needed.

To monitor multicollinearity, we calculated VIF values for the preliminary and intermediary models, flagging predictors with VIF > 5. High VIF values suggested inflated coefficient variance, potentially leading to unreliable conclusions on the significance of the model. Multicollinearity could be addressed by either collecting more data to boost the sample variations or respecifying the model.

With conditions and assumptions addressed, significance tests guided model refinement. After choosing a significance level, ANOVA assessed whether at least one predictor was significant, followed by T-tests to identify insignificant predictors, which were excluded to create reduced models. Partial F-tests confirmed whether reduced models retained explanatory power. The process aimed to refine a model where all predictors were significant; we would deploy that as our final model.

We then used automated methods, including All Possible Subsets, Forward, Backward, and Stepwise Selection, which optimized adjusted R^2 and AIC. We hoped to reach some agreement between automated and manual selections to validate the final model.

We identified leverage points, outliers, and influential data points in the final model to understand their impact. Leverage points were identified by their distance from the predictor center, and standardized residuals flagged outliers. Cook's Distance, DFFITS, and DFBETAS quantified data points' influence on fitted values and coefficients, informing their impact on the model. Normally no action would be pursued for the problematic observations unless there was a specific context that required us to exclude them.

Finally, we confirmed the final model met all assumptions, with minimal multicollinearity checked via VIF. This ensured confidence in making interpretations and answering the research question with the final model.

Results

Model Diagnostics

The scatterplot of the Response vs. Fitted Values showed a random scatter around the diagonal, indicating no systematic patterns and Condition 1 was satisfied. However Condition 2 appeared violated, as the pairwise scatterplot between AlcoholYear and BMI revealed a parabolic relationship.

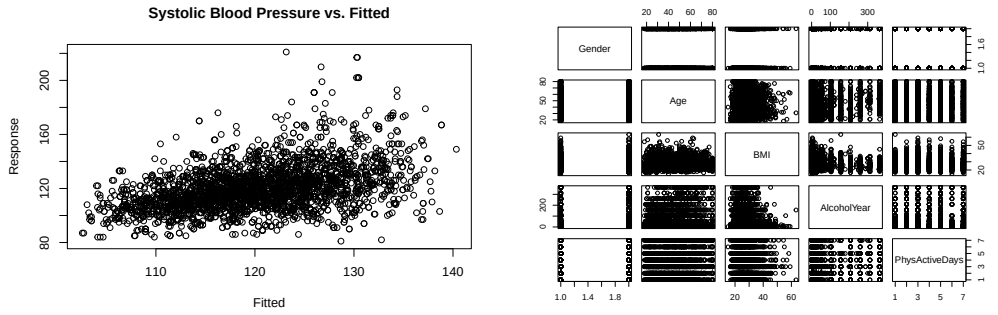


Figure 1: Response vs. Fitted plot (left) and pairwise predictor plot (right) for the preliminary model

There were violations for all model assumptions. Strong fanning and non-linear patterns on Residuals vs. BMI concerned the constant variance and linearity assumption. The stark deviation from the diagonal line on the Q-Q plot warned non-normality. Residuals vs. AlcoholYear showed serious issues with uncorrelated errors: Although AlcoholYear was a discrete predictor, the residual plot should still show random scatter among integers between 0 to 365, not systematic accumulations on certain values, which was shown here.

There was minimal issue with multicollinearity: The VIF for each predictor was between 1.01-1.05.

Model Transformations

Despite violation on Condition 2, we proceeded to correct the aforementioned model assumption violations first. A Box-Cox transformation was applied to both the response and predictors. Specifically, the response variable, BPSysAve, was transformed using an inverse transformation, Age was transformed using a square

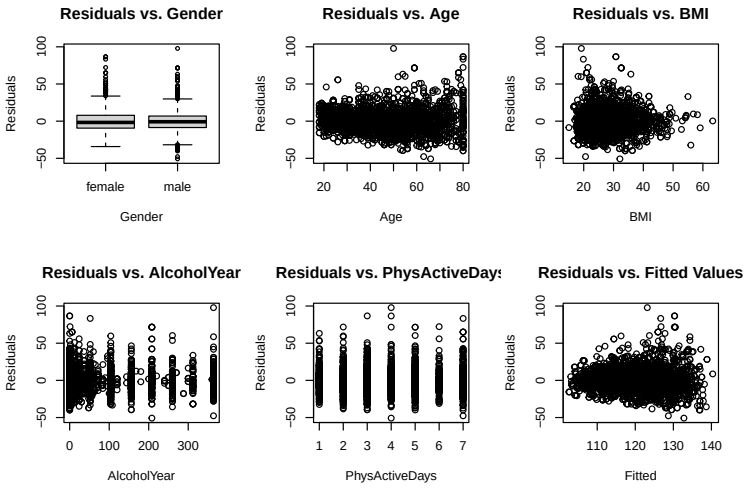


Figure 2: Residuals vs. Predictors in the preliminary model

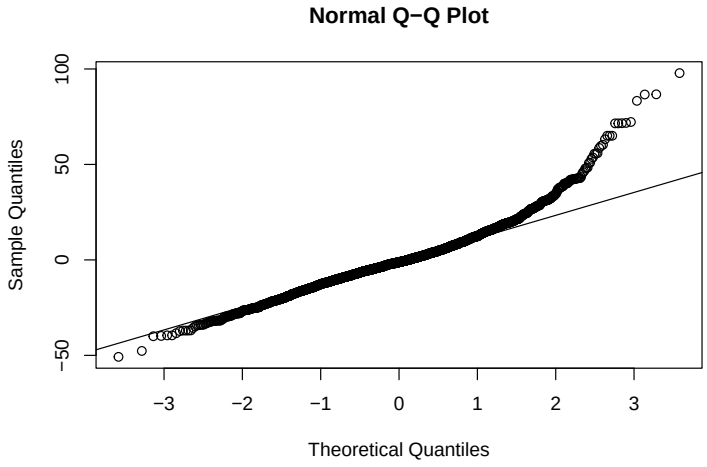


Figure 3: Q-Q plot for the preliminary model

root transformation; BMI was transformed using an inverse square root transformation; AlcoholYear was increased by one to avoid zero-entries, then log-transformed; PhysActiveDays was transformed using a square root transformation. The transformed model yieled

$$\text{BPSysAve}^{-1} = \beta_0 + \beta_1 \text{Gender} + \beta_2 \sqrt{\text{Age}} + \beta_3 \sqrt{\text{BMI}}^{-1} + \beta_4 \log(\text{AlcoholYear} + 1) + \beta_5 \sqrt{\text{PhysActiveDays}} + \epsilon$$

The transformation significantly corrected the violation of linearity and normality. It also improved constant variance for BMI and Condition 2 for AlcoholYear and BMI.

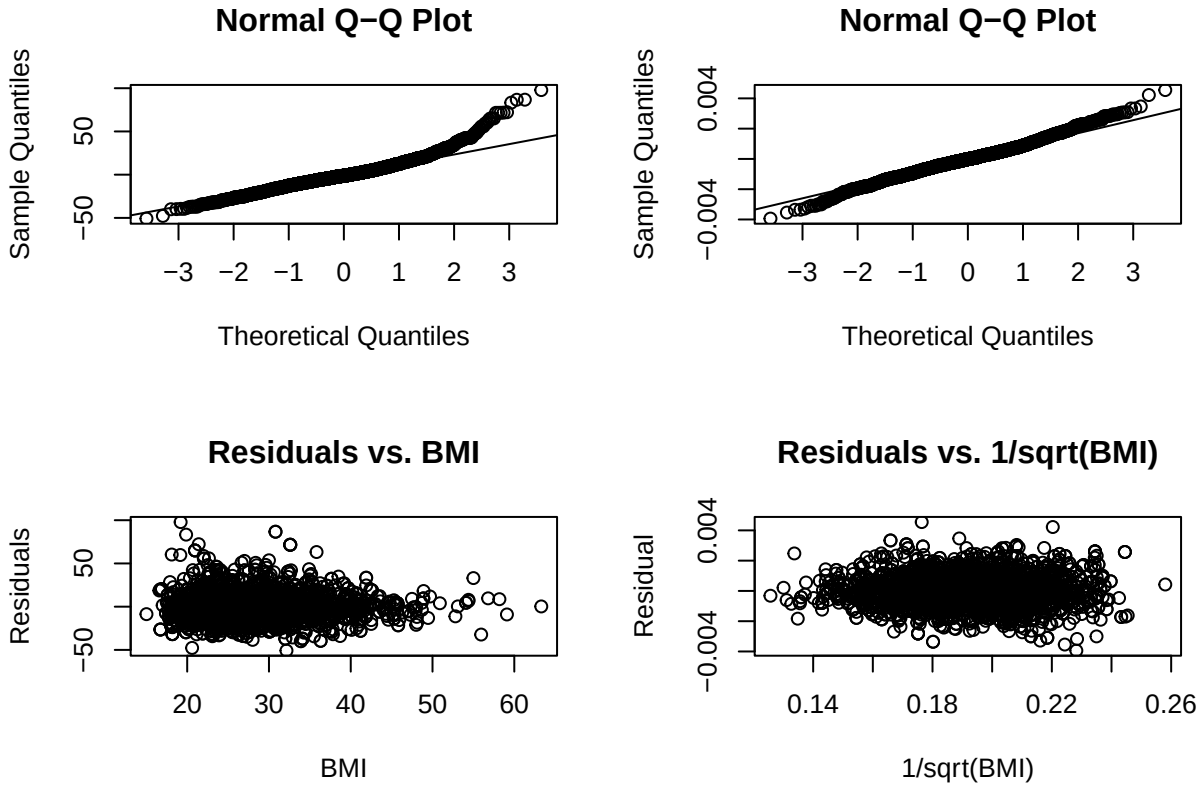


Figure 4: Before (left) and after (right) Box-Cox transformation, normality and constant variance improved

Manual Model Selection

We chose our significance level to be 0.05. The ANOVA overall significance test yielded a p-value of $2.2 \times 10^{-16} < 0.05$. This showed at least one predictor in the transformed model had a significant linear relationship with $BPSysAve^{-1}$. The p-values from the T-test showed that Gender , $\sqrt{\text{Age}}$, $\sqrt{\text{BMI}}^{-1}$, and $\log(\text{AlcoholYear} + 1)$ were statistically significant predictors. In contrast, the p-value of $\sqrt{\text{PhysActiveDays}}$ (0.635) exceeded the significance threshold, indicating that $\sqrt{\text{PhysActiveDays}}$ was not a statistically significant predictor.

We considered a simpler model that excludes $\sqrt{\text{PhysActiveDays}}$. A partial F-test was performed to evaluate whether the exclusion statistically compromises the explanatory power of the model. The F-value (0.228) exceeds 0.05, confirming that $\sqrt{\text{PhysActiveDays}}$ does not statistically improve the model and can be excluded.

The model suggested by the significance tests included the predictors: Gender, $\sqrt{\text{Age}}$, $\sqrt{\text{BMI}}^{-1}$, and $\log(\text{AlcoholYear} + 1)$, with BPSysAve^{-1} as the response variable. Now all predictors in the final model were statistically significant with BPV.

Automated Model Selection

We cross-validated our manually selected final model with automated selection methods. All Possible Subsets, Forward, Backward, and Stepwise automated selection were deployed. All four techniques offered the same final model as the manually selected model. Therefore we were confident that the model selected in the partial F-test was optimal.

Final Model

The final model was fitted with Gender, $\sqrt{\text{Age}}$, $\sqrt{\text{BMI}}^{-1}$, and $\log(\text{AlcoholYear} + 1)$. It gave the following regression equation:

$$\text{BPSysAve}^{-1} = 8.728\times10^{-3} - 3.845\times10^{-4}\text{Gender} - 2.905\times10^{-4}\sqrt{\text{Age}} + 1.019\times10^{-2}\sqrt{\text{BMI}}^{-1} - 2.475\times10^{-5}\log(\text{AlcoholYear}+1)$$

Diagnostics of Final Model

The final model reasonably satisfied all linear regression conditions and assumptions except showcasing correlated errors (see Figure 6-9 in Appendix). We left this as a limitation. The VIF values for all predictors in the final model were between 1.01-1.04, confirming no severe multicollinearity.

There were no leverage points, no influential points affecting all fitted values, and no influential points affecting their own fitted values. However, there were 3 outliers and 23 points that influenced all coefficients. We had no evidence to remove them.

Conclusions and Limitations

The final model reveals that Gender, Age, BMI, and alcohol drinking frequency, under transformation, have statistically significant combined effects on the inverse of BPSysAve, which directly relates to BPV. The coefficients quantify their respective influences. For instance, the coefficient of $\sqrt{\text{Age}}$ indicates that, holding other predictors constant, a one-unit increase in $\sqrt{\text{Age}}$ results in an expected change of -2.906×10^{-4} in the average of BPSysAve^{-1}. Overall, the model suggests that increases in $\sqrt{\text{Age}}$, $\sqrt{\text{BMI}}^{-1}$, and $\log(\text{AlcoholYear} + 1)$ are associated with a decreased expected average of BPSysAve^{-1}. Additionally, males with the same values for other predictors tend to have a lower expected BPSysAve^{-1}.

These findings mostly align with prior literature, reinforcing conclusions that male gender, older age, higher BMI, and greater alcohol consumption frequency are associated with increased BPV. Although PhysActiveDays– a predictor supported by literature– was removed during model selection, we suspected it was due to model assumption violations.

The model's significance lies in its ability to capture not only physiological influences on BPV but also the impact of lifestyle choices. This highlights the idea that while physiological factors may be harder to alter, lifestyle changes can meaningfully impact BPV with effort and determination.

However, the model has notable limitations. Its adjusted R-squared value of 0.1826 indicates that less than 20% of the variation in BPSysAve^{-1} is explained, limiting its ability to capture fluctuations in BPV. This weak explanatory power constrains insights into the drivers of BPSysAve^{-1} thereby BPV. Hence the information suggested by the final model might not be accurate.

Moreover, the model significantly violates the uncorrelated error assumption. The Residuals vs. log(AlcoholYear+ 1) plot shows strong clustering in vertical lines. Despite the discreteness of log(AlcoholYear + 1), random scatter among integers from 0-365 values is expected, not systematic accumulations on certain values. This violation indicates connections among data points in the population, contradicting the assumption of independent errors in linear regression. Consequently, predictor variance may be over- or underestimated, leading to unreliable standard error estimates in our final model.

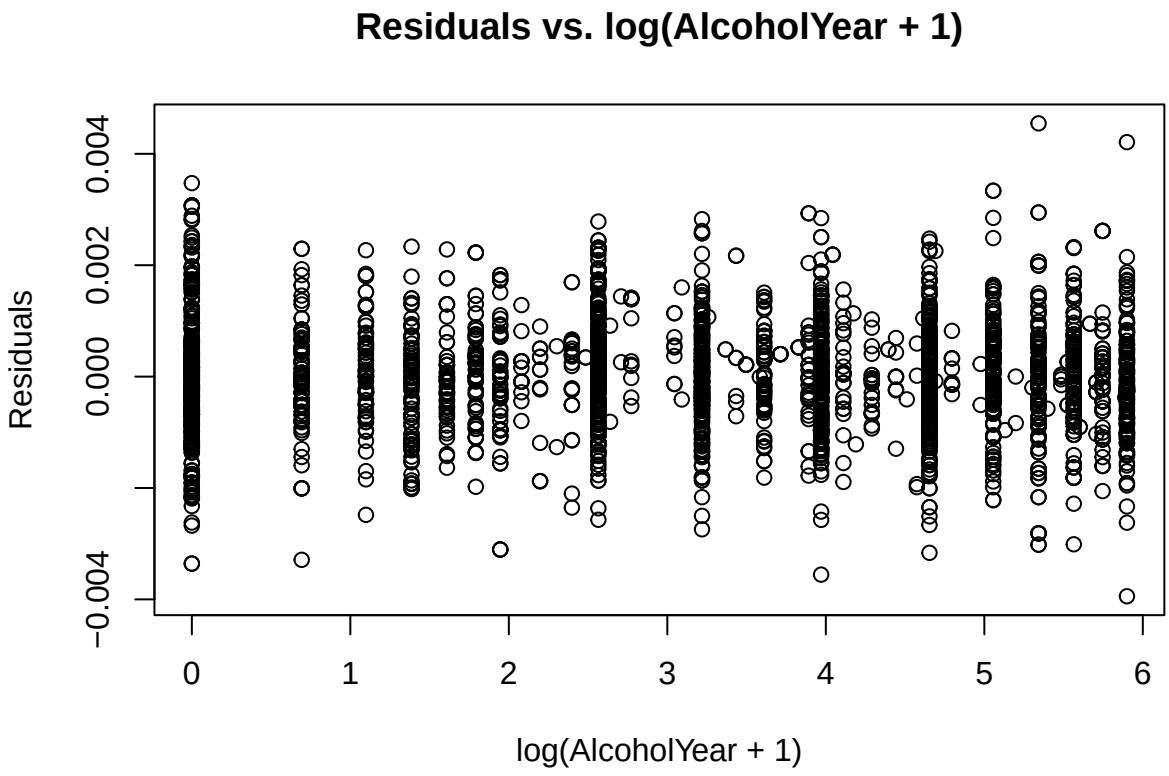


Figure 5: Residuals vs. log(AlcoholYear + 1) in the final model

Ethics Discussion

We employed manual selection methods to determine our final model, subsequently validating it using automated selection methods to ensure the choice was reasonable. Automated selection was not used as a starting point because our research question required a model that incorporated both physiological and lifestyle predictors. Since lifestyle predictors appeared less significant prior to model selection, we were concerned that an automated process might exclude them entirely. However, our manual selection approach—using T-tests and partial F-tests—produced the same outcome as all four automated methods we tested (all-possible-subsets, forward, backward, and stepwise selection), and all included the lifestyle factor log(AlcoholYear+1).

Ethically, we believe that in our specific context manual and automated methods are comparable. A small model with only five predictors like ours allows for detailed supervision, interpretation, and assessment at each step of the automated process, ensuring transparency and accountability. The primary ethical concern with automated selection lies in the potential for unintended negligence. Limited knowledge or control over the selection process can result in biased or unjust models being selected and deployed, potentially causing

harm. However, the close oversight we maintained transformed the automated methods from a potential "black box" into a transparent process, so negligence, whether intended or not, was avoided.

While automated selection carries risks of biased outcomes, our careful monitoring mitigated these risks, making it as reliable and effective as manual selection in this instance. That said, the ethical implications of manual versus automated selection methods may be very different in other contexts.

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Appendix

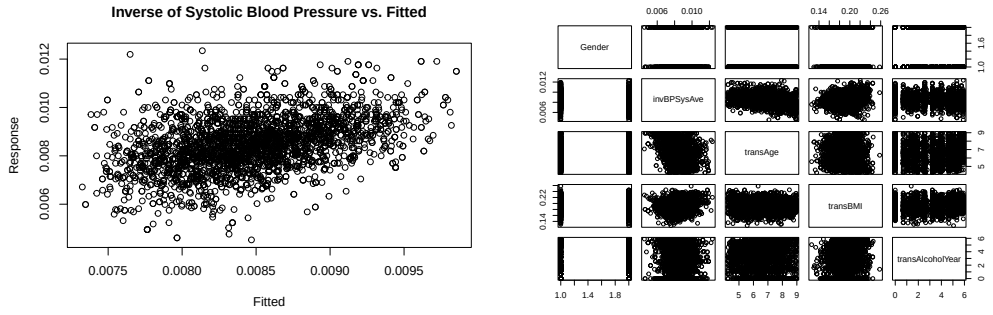


Figure 6: Condition 1 and 2 for the final model

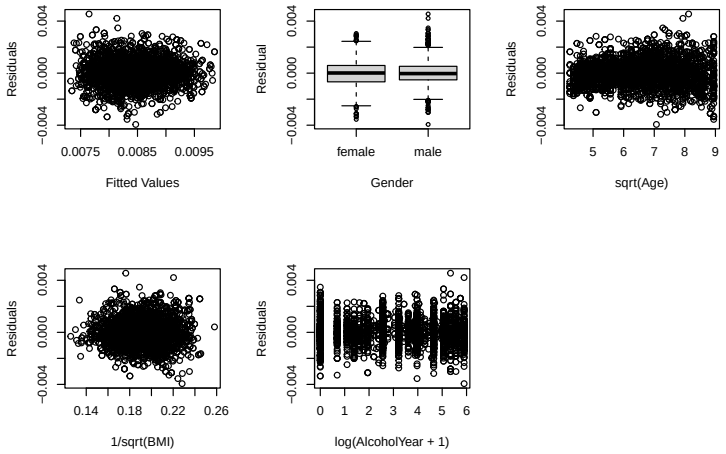


Figure 7: Residuals vs. predictors in the final model

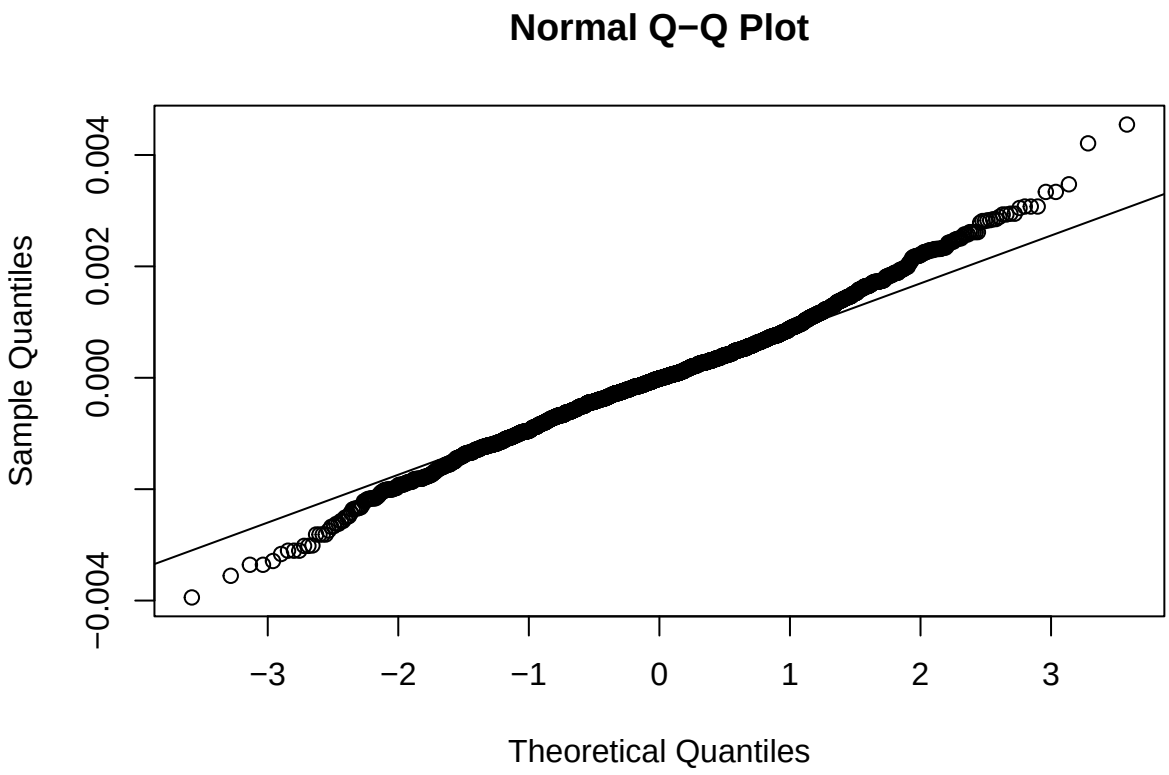


Figure 8: Q-Q plot for the final model